HÁSKÓLI ÍSLANDS

Computer Science - Computer Graphic TOL105M
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Verkefni 2

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In this programming project, you are to extend the viewpoints program so that it shows an entertaining small town with various additional features. Below are the new features:

- Modify the houses in the town so that they are of different sizes and shapes. Also, use more colors on the houses. You can have 3–4 types of houses instead of just one type as it is now.

- Include at least two cars that drive in opposite directions on the road (on separate lanes!).

- There must be a walkway/bridge that the cars drive through, i.e., some kind of building that goes over the road, but with an opening where the cars pass through. You may build this out of cubes!

- Add one new viewpoint (number 0), where the user is walking on the ground. Mouse movement to the side changes the user's current direction (and viewing angle), and the keys W-A-S-D move the user (W moves forward in the current direction, S moves backward, A moves left without changing direction, and D moves right). It makes sense to add variables that store the user's current position (in (x, y)) and current direction (also (x, y)). There is no need to implement collision detection here.

- There should be an airplane that circles above the town following a path shaped like the number 8 (a Lemniscate). There exists a version of this path that has a simple parametric representation: the Eight Curve. In it, it is also easy to find what direction the plane should be at each point in time. The airplane can be fairly simple: fuselage, wings, tail. Likely, you can manage with 4 cubes, or your own airplane model.

You should add these features one by one. They vary in difficulty. You may also use your creativity to add other interesting things.

https://tranquil-bombolone-907ebe.netlify.app/angel/viewpoints

Overview

I rewrote the viewpoints to make the current city, which has a bridge, sky scrapers, house and a roundabout with 4 cars, or 4 lanes, there is no collision logic. The houses are of different colors and so is the cars, the sky scrapers are different in their "window" as well as their own sizes. The airplane is circling over the city and user can go up or down and have a city view.i was using the initshaders and the color shader for this project.

Houses

The "houses" are made from scales cubes, the roofs are also cubes, I wanted to do a slanted roof but it was taking a bit too much time to be able to make it fit the houses and I stopped to focus on core functionalities as i was running out of time. The houses are of varying colors and shapes, wide, tall, rectangular and default.

Cars

There are now 4 cars in the lane with pos = (R * sin 0, R * cos 0) with lanes offset perpendicular to the radius, they are going opposite directions, i was going to put the lanes such that each car would be in each lane with a white line indicating the lanes but that was too much work and I couldn't finish on time. None however hit eachother or looks like they are hitting each other, therefore, there are no collision logic available.

Bridge

Made a cute wooden bridge that looks like a semi moon shaped curve that looks like is a woods stuck together, the cars can pass through it and it looks okay on the viewpoint.

Viewpoint

User can use the up and down buttons to go over the city and under the city.

Airplane

Airplane is flying over the city with a 8 shaped pattern with a x = A * sin(t), y = B * sin(2t) curve, plane is made of cubes consisting of its body, wings and tails.